Personal details and date of CV

- Surname: Mao
- First name: Zhu
- ORCID: 0000-0002-8464-2558
- Date of CV: Oct. 2026

Degrees

- PhD in Photogrammetry and Remote Sensing, LIESMARS, Wuhan University, Wuhan, Jun. 2023
- MSc in Surveying and Mapping Engineering, LIESMARS, Wuhan University, Wuhan, Jun. 2018
- BSc in Geographic Information Science, Nanjing University of Information Science & Technology (NUIST), Nanjing, Jun. 2016

Current employment

- Postdoctoral researcher, Department of Forest Sciences, University of Helsinki (research group ForTech), Jan. 2024–present
  - Stage of academic research career: Stage II
  - Researching artificial intelligence (AI) in forestry, photogrammetry and remote sensing, with a focus on GeoAI for precision forestry. The broader goal of my research is to advance the scientific understanding of boreal forest structure and processes through GeoAI, and to translate this understanding into methods supporting precision forest operations, bridging AI, remote sensing and photogrammetry with forest operations and ecology. This interdisciplinary trajectory directly underpins the GeoAI approach of the current research proposal.

Previous work experience

- Editorial Assistant (part-time), *Geo-spatial Information Science* (GSIS), Wuhan, Jan. 2021–Dec. 2023
  - Open-access international journal founded by Wuhan University and published with Taylor & Francis; scope covers photogrammetry, remote sensing, GIS, and geomatics.
- Doctoral researcher, LIESMARS, Wuhan University, Wuhan, Sep. 2019–Jun. 2023
  - Research on oblique photogrammetric 3D reconstruction and AI-based scene understanding.
- Business Analyst, IBM CIO Identity team, Chengdu, Jul. 2018–Aug. 2019
  - Worked in an international team on enterprise identity and access management.

Research funding and grants

- GeoAISoil, funded by private foundations, May 2025–Apr. 2027
  - Role: key researcher (PI: Prof. J. Uusitalo)
  - In collaboration with Arbonaut
- Precision Forest Harvesting, Jan. 2024–Jun. 2025
  - Role: key researcher (PI: Prof. J. Uusitalo)

Research output

- Publications: 14 peer-reviewed journal articles on AI-based remote sensing and photogrammetry, with applications in forestry and geoinformation science. (5 as first author, 1 corresponding author).
- Open-source software

  - MT-GAT: multi-task learning graph attention network for predicting soil organic layer thickness and organic matter in boreal forests, 2026
  - LTA-Logging Trail Analyzer: a Python-based workflow for deriving and evaluating logging trail networks from forest-machine GNSS data, 2026
  - SR-Supporting-ITD: individual tree detection and canopy stratification from half-metre aerial data, 2025
- **Open datasets**
  - URFD: slender urban road facility extraction from oblique aerial images, 2023
  - GFSD: glass facade segmentation from oblique aerial images, 2022

Research supervision and leadership experience

- **Co-supervisor** of one doctoral researcher, University of Helsinki

Teaching merits

- **Guest lecturer**, MSc courses in Forest Sciences, University of Helsinki, 2024–2026
  - FOR-265 *Terramechanics and Trafficability Prediction*, 2026
  - FOR-263 *Operations Research in Forest Resource Management*, lecture "Computer Vision in Forestry", 2024, 2026
  - FOR-280 *Precision Harvesting*, 2025
  - FOR-018 *Introduction to the MSc Programme in Forest Sciences*, 2025, 2026

Other key academic merits

- Lead Guest Editor, Special Issue "Remote Sensing of Individual Trees: Delineation, Characterization, and Applications", *Remote Sensing* (MDPI), 2026
- Peer reviewer for international journals:
  - including *ISPRS Journal of Photogrammetry and Remote Sensing* (ISPRS P&RS), *International Journal of Applied Earth Observation and Geoinformation* (JAG), *Scientific Data* (SD), *Remote Sensing* (RS) and *Geo-spatial Information Science* (GSIS)
- Oral presentations
  - "Mapping tree-level structural diversity for precision harvesting operations: A comparison of open-source and drone datasets", Forest Science Day 2025, Helsinki, Nov. 2025
  - "Deep-learning-based individual tree detection for tomorrow's precision harvesting operations", FORMEC 2025, Joensuu, Jun. 2025

Scientific and societal impact

- Promoting open science: I openly release the code and data behind my research. The methods MT-GAT and SR-Supporting-ITD and the benchmark datasets URFD and GFSD are publicly available on GitHub (see Research output), allowing other researchers to reproduce the results and build on them.
- Utilising research outputs: In the GeoAISoil project, I collaborate with Arbonaut, a company providing geospatial and forest inventory services, to transfer GeoAI-based soil property mapping toward practical forest planning and operations.

Skills and Languages

- **Programming:** Python (PyTorch, scikit-learn, GDAL)
- **Software:** ArcGIS, QGIS, CloudCompare, Bentley ContextCapture, LaTeX
- **Computing environments:** Windows, Linux (Ubuntu), CSC supercomputing, Git
- **Languages:** Chinese (native), English (fluent)